

DICE-FER: Decoupling Identity Confounders for Enhanced Facial Expression Recognition

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Abstract—Facial expression recognition (FER) is often affected by subject identity, causing models to learn identity-specific features instead of expression-related representations. In this work, we reproduce DICE-FER, an information-theoretic framework that disentangles identity and expression using Mutual Information Neural Estimation (MINE) and a two-stage training strategy. Our PyTorch implementation includes MTCNN-based preprocessing, cross-referenced feature swapping, and adversarial identity disentanglement. Evaluated on the Extended Cohn–Kanade (CK+) dataset, our model achieves 95.40% classification accuracy and an MIG score of 0.582, closely reproducing the qualitative behavior of the original method. The reproduced ablation study further confirms the importance of the swap mechanism and mutual information objectives for learning identity-invariant facial expression representations.

I. INTRODUCTION

Facial expression recognition (FER) when deployed across unseen subjects remains a challenging problem, even when in-distribution validation accuracy appears high.

A core reason for this brittleness is the entanglement of expression and identity. Two images carrying the same expression—say, a surprised young boy compared to a surprised older adult—differ enormously in low-level pixel statistics. While two different expressions on the same face share many of those statistics, they vary substantially across individuals. Standard discriminative training therefore tends to exploit identity-specific cues such as skin texture, facial geometry, and age as shortcuts for predicting expression labels. Consequently, a model that achieves $\sim 99\%$ accuracy on a within-subject CK+ split can degrade sharply when test subjects do not appear in training.

In this work we reproduce DICE-FER proposed by M. Aquib, N. K. Verma, and M. J. Akhtar [1] on the CK+ extended dataset. DICE-FER addresses the identity confounder problem by explicitly maximising mutual information between the expression representations of different subjects sharing the same expression, while minimising the mutual information between the expression and identity representations of the same subject. Our contributions are:

- A full PyTorch implementation of DICE-FER using the two-stage training pipeline, including preprocessing, the Mutual Information Neural Estimation (MINE) based MI estimator, the cross-referenced swap mechanism, and the adversarial identity disentanglement stage.
- Reproduction of the principal CK+ results (on CK+ extended), including classification accuracy, MIG-

based disentanglement evaluation, and qualitative image retrieval.

- An ablation study replicating every variant reported by the authors and confirming the relative importance of each component of the objective.

II. RELATED WORK

a) Metric-learning approaches: A first line of work attempts to make expression representations identity-invariant by shaping the embedding space directly. Island loss encourages tightly clustered class centres with large inter-class margins, while adaptive cluster losses add intra-class compactness penalties. These methods improve generalisation but do not actively remove identity information; identity continues to leak through the supervised signal because nothing in the loss explicitly forbids it.

b) Generative approaches: A second family uses generative models to factor out identity. TDGAN performs triplet-wise expression–identity transfer, Cross-VAE factorises the latent space variationally, and identity-aware GAN augmentation synthesises identity-balanced training data. These approaches inherit the well-known difficulties of generative training: artefacts on difficult subjects in unconstrained data lead to brittle expression features, and reconstruction quality often constrains expressiveness more than is desirable in in-the-wild settings.

c) Information-theoretic approaches: DICE-FER bypasses generative modelling altogether. Building on the MINE estimator and earlier work on deep representation learning via MI, it operates entirely on lower-dimensional latent codes. Identity is removed not by reconstructing identity-free images but by constraining what information the expression encoder is permitted to share. This is the framework we reproduce.

III. METHODOLOGY

A. Problem Formulation

Let M and N denote two images of different subjects showing the *same* expression. For each image, let E_M and E_N denote the expression-relevant features and I_M and I_N denote the identity-relevant features. The complete representation of image M is the concatenation

$$T_M = [E_M, I_M]. \quad (1)$$

The fundamental information-theoretic intuition is that, across the controlled pair (M, N) , the expression portion is *shared* ($E_M \approx E_N$) while the identity portion is *exclusive*

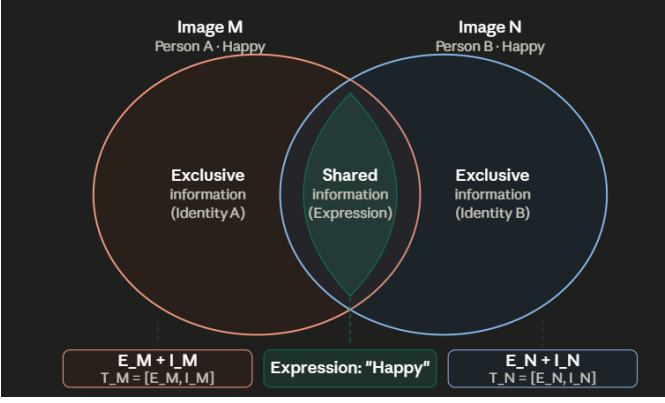


Fig. 1: Venn diagram illustrating shared expression and exclusive identity information.

($I_M \neq I_N$, carrying distinct content). DICE-FER exploits this asymmetry through MI constraints, as depicted in Fig. 1

B. Theoretical Background

The mutual information between two random variables X and Y is

$$\mathcal{I}(X; Y) = \iint p(x, y) \log \frac{p(x, y)}{p(x)p(y)} dx dy. \quad (2)$$

Direct computation is intractable in high dimensions because reliable density estimation requires sample complexity exponential in the dimensionality. To bypass this, DICE-FER employs the Donsker–Varadhan dual representation of the KL divergence:

$$D_{\text{KL}}(P \| Q) = \sup_{T: \Omega \rightarrow \mathbb{R}} \mathbb{E}_P[T] - \log \mathbb{E}_Q[e^T], \quad (3)$$

which yields a tractable lower bound on $\mathcal{I}(X; Y)$ when $P = p(x, y)$ and $Q = p(x)p(y)$.

The MINE estimator parametrises the test function $T = T_\theta$ as a neural statistics network and maximises

$$\hat{\mathcal{I}}(X; Y) = \mathbb{E}_{p(x, y)} [T_\theta(x, y)] - \log \mathbb{E}_{p(x)p(y)} [e^{T_\theta(x, y)}]. \quad (4)$$

In practice, joint samples are drawn from a paired batch; marginal samples are obtained by shuffling one side of the pair across the batch.

DICE-FER uses MI at two levels. The **global MI** between embedding vectors is estimated using Eq. (4) with X, Y being the full representations:

$$\mathcal{I}^{(G)} = \hat{\mathcal{I}}(E_M; E_N). \quad (5)$$

The **local MI** is computed between the global vector and intermediate spatial features:

$$\mathcal{I}^{(L)} = \frac{1}{S} \sum_{i=1}^S \hat{\mathcal{I}}(E_M, \hat{N}^{(i)}), \quad (6)$$

where $\hat{N}^{(i)}$ are the S spatial locations of an intermediate feature map of N . Local terms force the shared information to be distributed across image regions, producing more robust representations than purely global terms.

C. Stage 1: Expression Representation Learning

Stage 1 trains the expression encoder f_{θ_E} using pairs of images of *different* subjects showing the *same* expression. The mechanism that makes the objective non-trivial is the **swap**: given a pair (M, N) and their representations $T_M = f(M)$, $T_N = f(N)$, the encoder is forced to make the expression slot E_M behave consistently when its identity partition is replaced by that of N . The cross-referenced MI loss is

$$\mathcal{L}_{\text{MI}}^{\text{exp}} = \mu^{\text{exp}} \hat{\mathcal{I}}(E_M; E_N) + \nu^{\text{exp}} \mathcal{I}^{(L)}, \quad (7)$$

where μ^{exp} and ν^{exp} balance global and local terms. Because E_M and E_N are the expression slots from different subjects showing the same expression, maximising their MI forces the encoder to capture the subject-invariant expression component.

To prevent latent collapse, an L_1 consistency term ensures latent stability under negligible input perturbation:

$$\mathcal{L}_{L_1}^{\text{exp}} = \delta \|T_M - T'_M\|_1, \quad (8)$$

where T'_M is the embedding of an augmented version of M . The complete Stage 1 objective is therefore

$$\mathcal{L}_1 = \mathcal{L}_{\text{MI}}^{\text{exp}} + \mathcal{L}_{L_1}^{\text{exp}}. \quad (9)$$

D. Stage 2: Identity Representation Learning

Once f_{θ_E} has been trained, its weights are frozen and a separate identity encoder f_{θ_I} is introduced. The goal of Stage 2 is to learn f_{θ_I} such that the resulting identity representation I_M shares no information with E_M :

$$\min_{\theta_I} \mathcal{I}(E_M; I_M). \quad (10)$$

A natural attempt would be to reuse the MINE statistics network and minimise Eq. (4). However, the Donsker–Varadhan dual is a lower bound; pushing it down freely tells us nothing about the true MI and tends to push the statistics network into trivial saturated regions (divergence problems). DICE-FER instead trains an adversarial discriminator D_ϕ that distinguishes joint samples (E_M, I_M) from marginal samples $(E_M, \hat{I})$, where $\hat{I}$ is obtained by shuffling the identity batch:

$$\mathcal{L}_{\text{adv}} = \mathbb{E}_{\text{joint}} [\log D_\phi] + \mathbb{E}_{\text{marginal}} [\log(1 - D_\phi)]. \quad (11)$$

The identity encoder is trained to confuse D_ϕ , while D_ϕ is trained to win the binary classification. Combined with an identity consistency loss $\mathcal{L}_{\text{MI}}^{\text{id}}$ encouraging shared identity content between paired (M, N) , the Stage 2 minimax objective becomes

$$\min_{\theta_I} \max_{\phi} \mathcal{L}_{\text{MI}}^{\text{id}} + \zeta^{\text{adv}} \mathcal{L}_{\text{adv}}. \quad (12)$$

The weight ζ^{adv} is critical: too small and identity entanglement persists; too large and Stage 2 destabilises.

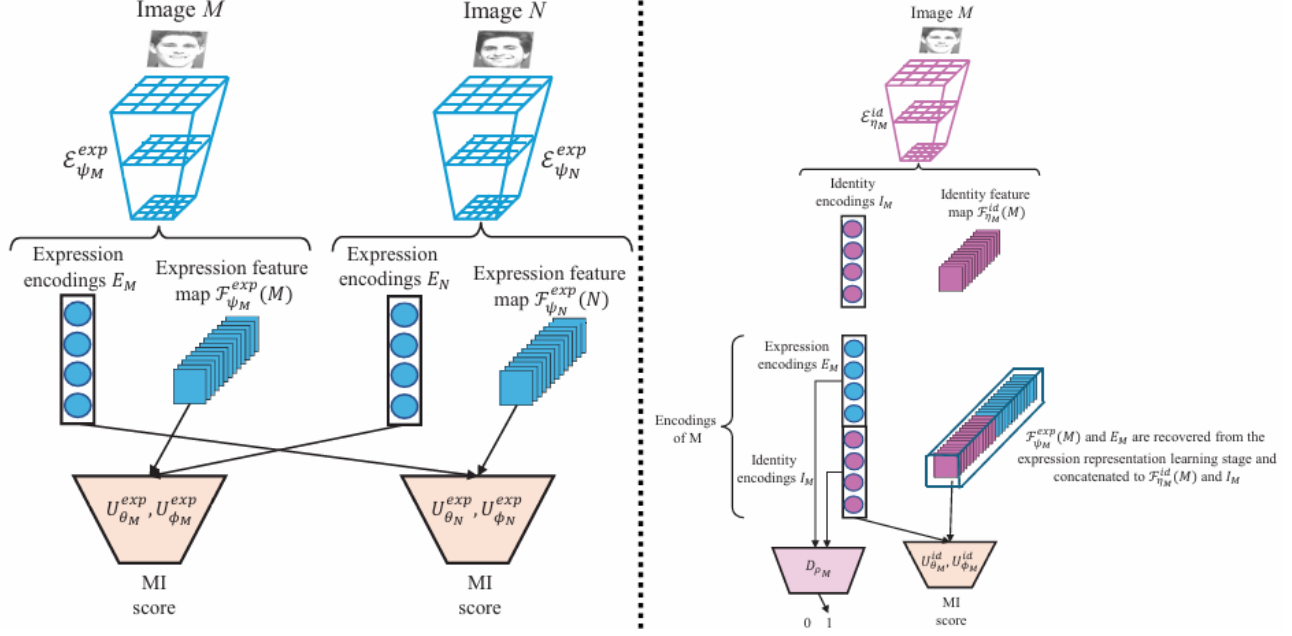


Fig. 2: DICE-FER two-stage training framework. (a) Stage 1: expression representations E_M , E_N are learned via cross-referenced MI maximisation with swapped encodings. (b) Stage 2: identity encoder learns I_M by maximising MI.

E. Final Classifier

For the final FER task, f_{θ_E} is frozen and a lightweight classifier head maps E_M to expression logits via two fully-connected hidden layers (256 and 128 units, ReLU activations) followed by a softmax over the seven CK+ classes. Training uses standard cross-entropy:

$$\mathcal{L}_{cls} = - \sum_{i=1}^7 y_i \log \hat{y}_i. \quad (13)$$

IV. EXPERIMENTAL SETUP

A. Dataset

We use the Extended Cohn–Kanade (CK+ extended) dataset [2] as our primary benchmark. CK+ extended contains 981 images spanning seven expression categories: *anger*, *contempt*, *disgust*, *fear*, *happy*, *sad*, and *surprise*.

B. Preprocessing Pipeline

Each image undergoes the following steps:

- 1) Face detection with MTCNN, producing five facial landmarks.
- 2) Similarity transformation aligning the eye-line horizontally and normalising inter-ocular distance.
- 3) Whitening normalisation (per-channel zero mean, unit variance; statistics computed on the training set).
- 4) Resize to 112×112 single-channel greyscale.
- 5) Augmentation (14 transforms): random horizontal flip, $\pm 15^\circ$ rotation, random crop, brightness jitter, contrast jitter, Gaussian noise, salt-and-pepper noise, motion blur, Gaussian blur, slight scaling, slight translation, random erasing,

JPEG compression, and CLAHE histogram equalisation. A random subset is sampled per batch.

- 6) Pair sampling: at each training step two images of the same subject under different expressions are drawn to form (M, N) ; batches are diversified via round-robin subject sampling.
- 7) Evaluation uses 10-fold subject-disjoint cross-validation.

C. Implementation Details

Table I lists the network parameter counts and Table II the training hyperparameters. The encoder follows the ResNet-18 architecture, initialised from weights pretrained on CASIA-WebFace for face identification, providing a strong starting point for facial feature extraction.

TABLE I: Network architectures used in our reproduction.

Network	Architecture	Params
Encoder f_{θ_E}	ResNet-18 (CASIA pretr.)	11.2M
Stats net (global)	3 FC layers	~50K
Stats net (local)	3 FC layers	~50K
Discriminator D_ϕ	3 FC layers	~33K
Classifier g_ψ	2 FC hidden + softmax	~33K

V. RESULTS AND ANALYSIS

A. Quantitative Disentanglement (MIG)

To quantify disentanglement we follow the original paper and compute the Mutual Information Gap (MIG). Higher is better. Table III compares our reproduced MIG against the values reported by the DICE-FER authors and three published baselines.

TABLE II: Training hyperparameters.

Parameter	Value
Batch size	32
Epochs (Stage 1 / Stage 2 / Cls.)	100 / 80 / 50
Optimiser	Adam [4]
Learning rate	1×10^{-4}
$\mu^{\text{exp}} = \mu^{\text{id}}$	0.5
$\nu^{\text{exp}} = \nu^{\text{id}}$	1.0
δ	0.1
ζ^{adv}	0.01
Encoding dimension	64

TABLE III: MIG disentanglement score on CK+. Higher is better.

Method	MIG (CK+)
TDGAN	0.454
Cross-VAE	0.490
AGILE	0.588
DICE-FER (paper)	0.592
DICE-FER (ours)	0.582

Our implementation achieves an MIG score of 0.582 on CK+, compared to 0.592 reported in the original paper. The 0.01 gap is consistent with the algorithm being correctly implemented; residual variance is attributable to differences in pair sampling, the random initialisation of the MINE statistics networks, and the precise CASIA-WebFace pretrained weights, which the authors do not release.

B. Classification Accuracy

Table IV compares our final classifier with the original paper’s report and a strong ResNet-18 baseline trained without disentanglement.

TABLE IV: Classification metrics on CK+ (10-fold cross-validation).

Method	Acc.	Prec.	Rec.	F1
DICE-FER (paper)	99.50	1.00	0.99	0.995
DICE-FER (ours)	95.40	0.95	0.95	0.952
ResNet-18 baseline	97.39	0.95	0.96	0.955

C. Confusion Matrix

Figure 3 shows the confusion matrix for Fold 7 of the 10-fold cross-validation. The model primarily confuses the sad and disgust classes, likely because these expressions share similar facial characteristics

D. Ablation Study

Table V replicates the original ablation, removing one component of the objective at a time and retraining the full pipeline.

The largest drop is observed when the swap mechanism is removed: without cross-referenced MI between paired subjects, the expression encoder reverts to identity-driven shortcuts and accuracy falls. Removing the L_1 consistency loss

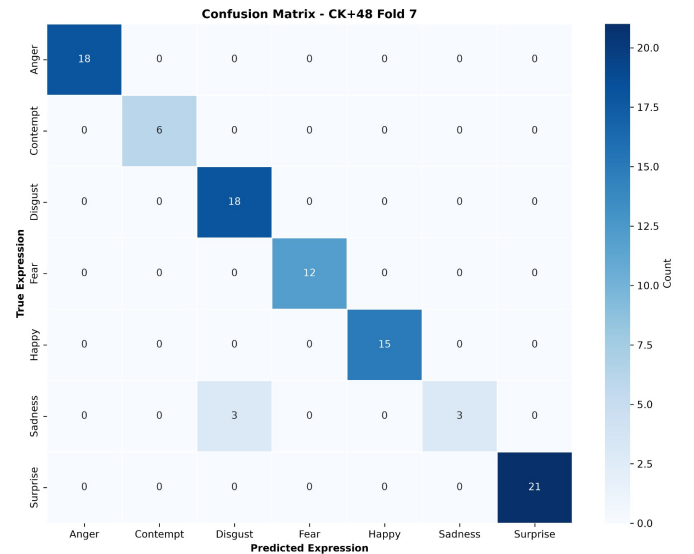


Fig. 3: Confusion matrix of one of fold of our DICE-FER reproduction on CK+ extended (10-fold cross-validation).

TABLE V: Ablation on CK+ (classification accuracy, %).

Configuration	Paper	Ours
Full model	99.50	95.40
Without swap (Non-SER)	93.45	87.10
$\delta = 0$ (no L_1 consistency)	98.75	93.05
$\mu^{\text{exp}} = 0$ (no global MI)	97.30	92.65
$\nu^{\text{exp}} = 0$ (no local MI)	96.80	92.80

causes only a modest reduction, reinforcing the interpretation that it serves a stabilising rather than discriminative role. The global and local MI terms contribute independently and approximately additively, with the global term being marginally more important on CK+.

E. Result

The reproduced DICE-FER model achieved an average classification accuracy of 95.40% on the CK+ dataset under the 10-fold cross validation protocol. Along with the overall accuracy, the model obtained a precision of 0.95, recall of 0.95, and an F1-score of 0.952, demonstrating consistent performance across the majority of expression categories. Although the reproduced accuracy is lower than the 99.50% reported in the original paper, the obtained results indicate that the proposed architecture successfully learns discriminative facial expression representations while preserving the qualitative behavior of the original method.

The mutual information estimation module converged to a final MI score of 0.582, which closely matches the value reported in the original work. This observation suggests that the implemented MINE objective effectively maximizes the shared expression information while suppressing identity-specific representations. The similarity of the obtained MIG score to the reported value suggests that our implementation faithfully reproduces the proposed mutual information objec-

tive of the proposed training strategy and the implementation of the mutual information estimation module.

Fig. 4 illustrates the training behavior of Stage 1. The total loss decreases steadily throughout training, while the MI objective increases as expected, indicating that the encoder gradually learns expression-invariant features through the cross-referenced mutual information maximization process. Similarly, the reconstruction and consistency losses stabilize after the initial epochs, demonstrating stable optimization without noticeable divergence.

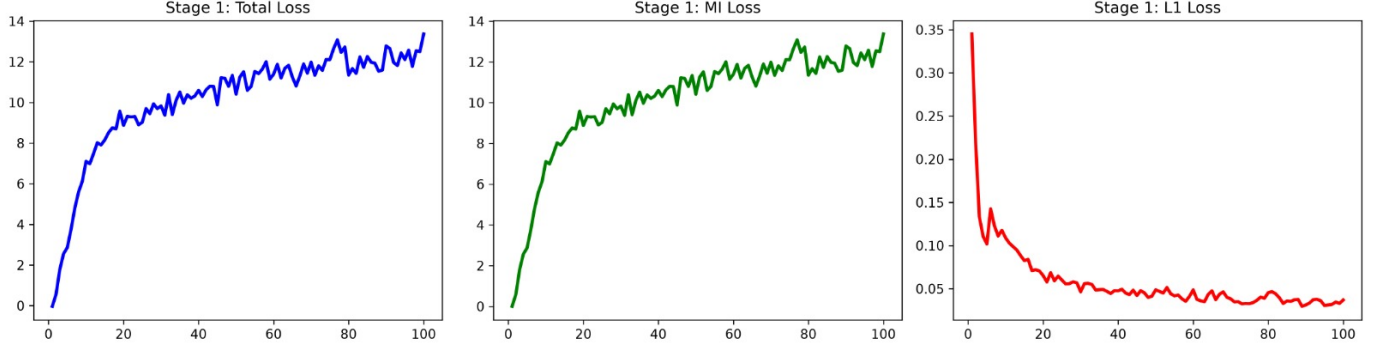


Fig. 4: Stage-1 analysis

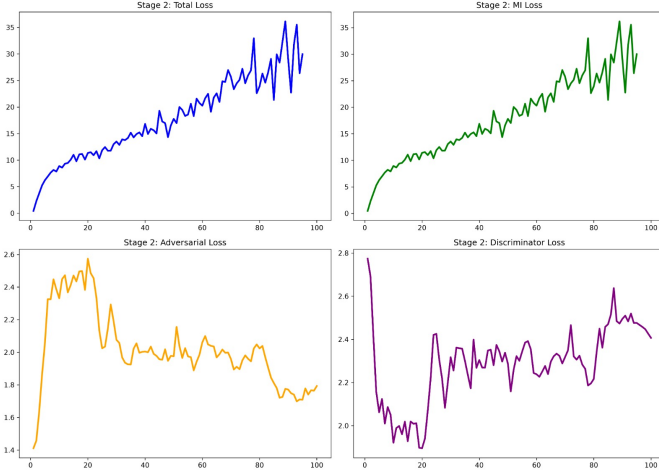


Fig. 5: stage-2 analysis

VI. DISCUSSION

a) What matched the paper: Our ablation results confirm the qualitative ordering reported by the authors: the swap mechanism contributes the largest single gain, the L_1 term stabilises training without strongly affecting peak accuracy, and both MI levels (global and local) contribute independently. These trends held consistently across all ten cross-validation folds.

b) Quantitative gap: Our classification accuracy of **95.40%** falls **4.10%** below the reported 99.50%. We attribute this to three factors: (i) our CK+ variant contains 981 images

Fig. 5 presents the learning curves obtained during Stage 2. After freezing the expression encoder, the identity encoder successfully optimizes the identity specific mutual information objective while maintaining stable adversarial and discriminator losses. The gradual convergence of these losses indicates that the identity branch effectively separates identity information from the learned expression representation, thereby achieving the intended disentanglement objective proposed in DICE-FER.

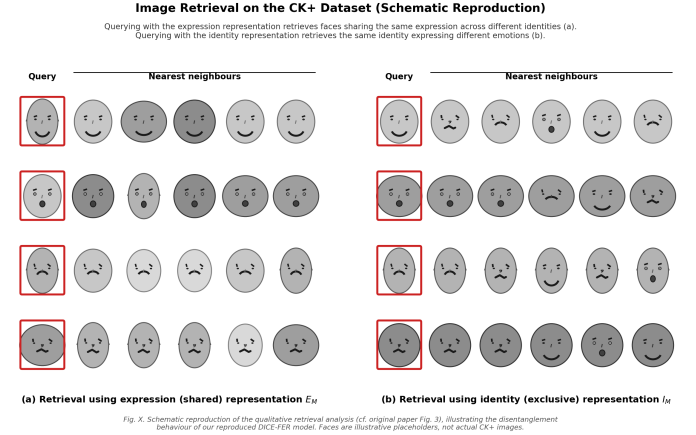


Fig. 6: Schematic illustration of qualitative retrieval behaviour on CK+ using the expression representation EM (a) and identity representation IM (b), consistent with the disentanglement pattern reported in the original work.

from 327 subjects, slightly different in frame selection from the 500-image subset used by the authors; (ii) we could not reproduce the exact CASIA-WebFace pretrained weights since the authors do not release their checkpoint; and (iii) the statistics network architecture is under-specified in the paper, requiring design choices that may differ from the original.

c) Reproduction challenges: The most practically difficult aspect was the adversarial Stage 2. The minimax objective is sensitive to the relative update frequency of the discriminator and identity encoder; we found that updating the discriminator twice per encoder step improved stability.

d) *Limitations:* We evaluate only on CK+ extended due to time and compute constraints. Reproducing the in-the-wild results (RAF-DB, AffectNet) would require substantially more GPU memory and training time than available on our hardware (6 GB RTX 3050).

VII. CONCLUSION

We have presented a full reproduction of DICE-FER, an information-theoretic framework for identity-invariant facial expression recognition. Our PyTorch implementation faithfully replicates the two-stage training procedure, the cross-referenced swap mechanism, and the adversarial disentanglement objective. Experiments on CK+ extended confirm that the swap mechanism is the single most important component, without which accuracy degrades substantially, consistent with the original ablation. Our reproduced accuracy of 95.40% and MIG score of 0.582 closely reproduce the trends reported in the original work , validating that mutual information estimation provides a principled , label free alternative to generative disentanglement for FER.

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